



DC – AI – Routing – Switching – Security – Wireless – Automation

Power Observability

An HPE Juniper Networking Techpost

Nicolas Fevrier

2026-05-29

HPE



Contents

1	Juniper Routers and Switches Power Observability	2
1.1	Introduction	2
1.2	TL;DR	2
1.3	Principles and Vocabulary	4
1.4	Disclaimers and Methodology	5
1.5	Devices with No Counters For “Input Power”	6
1.6	MX204	6
1.7	MX240 Chassis	7
1.8	MX480 Chassis	7
1.9	MX960 Chassis	8
1.10	MX301	9
1.11	ACX7020	10
1.12	QFX5110/QFX5120/QFX5200/QFX5210	11
1.13	Devices with “Input Power” Counters	11
1.14	MX304	11
1.15	MX10008 Chassis	12
1.16	PTX10001-36MR	15
1.17	PTX10002-36QDD	17
1.18	PTX10000 Chassis	18
1.19	ACX7024	21
1.20	ACX7100	22
1.21	ACX73xx	23
1.22	ACX7500	24
1.23	QFX5130	25
1.24	QFX5220	26
1.25	QFX5230	28
1.26	QFX5240/5241	29
1.27	QFX5700	30
1.28	Conclusion	31
1.28.1	Useful links	31
1.28.2	Glossary	31
1.28.3	Acknowledgments	32

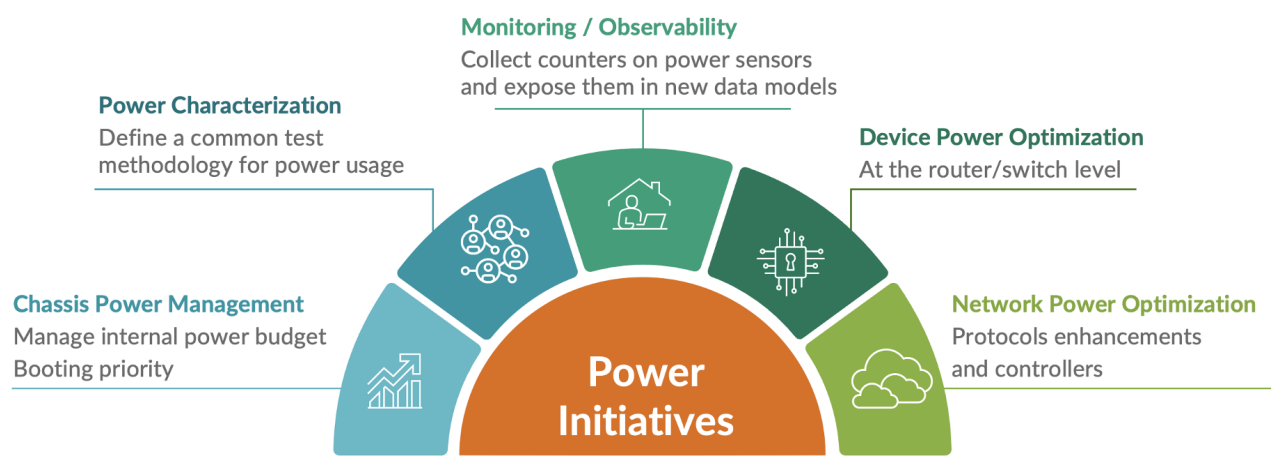
1 Juniper Routers and Switches Power Observability

Nicolas Fevrier - 05/29/2026

Your routers and switches display power usage counters, but what are they representing and can you trust them? Let’s try to answer these questions today.

1.1 Introduction

Different initiatives around power are ongoing, from characterization (power calculators) to network-level power optimization, but today we will focus on the observability part: what can we monitor and can we trust these numbers.



1.2 TL;DR

“It’s an awfully long article, I’m a busy engineer and don’t have time for all the details, just gimme the results please...”

Ok, ok... I admit it’s a very long article, so the following chart displays in a synthetic manner:

- Does the product offer a counter for Input Power (check the definition of input/output power in following section) and if it does support it, what is the CLI exposing this counter?
- When the counter is available, what’s the level of accuracy you can expect?

Product	Output Power	Input Power	Accuracy Delta of CLICompared to PDU
MX204(Junos)	show chassis env pem	Not Available	Not Applicable sinceInput is not available
MX240 Chassis(Junos)	show chassis powershow chassis env pem	Not Available	Not Applicable sinceInput is not available

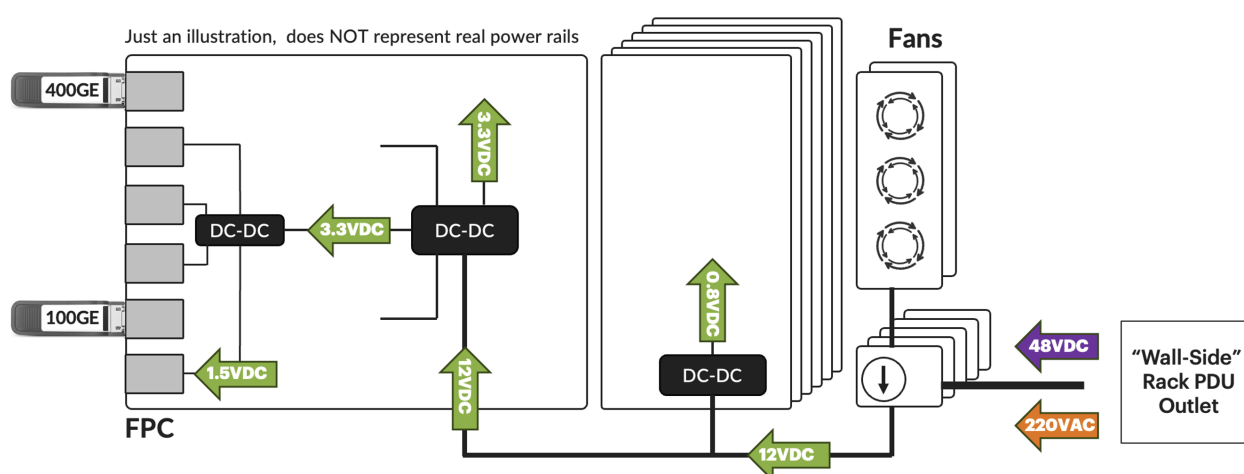
Product	Output Power	Input Power	Accuracy Delta of CLI Compared to PDU
MX480 Chassis(Junos)	show chassis powershow chassis env pem	Not Available	Not Applicable since Input is not available
MX960 Chassis(Junos)	show chassis powershow chassis env pem	Not Available	Not Applicable since Input is not available
MX301(Junos)	show chassis powershow chassis env pem	Not Available	Not Applicable since Input is not available
MX304(Junos)	show chassis powershow chassis env pem	show chassis env pem	Measured 2% deviation
MX10000 Chassis(Junos)	show chassis powershow chassis env pem	show chassis env pem	Measured from 2 to 5% deviation
PTX10001-36MR(Junos EVO)	show chassis power	show chassis power	Identical
PTX10002-36QDD(Junos EVO)	show chassis power	show chassis power	Identical
PTX10000 Chassis(Junos EVO)	show chassis power	show chassis power	Identical
ACX7020(Junos EVO)	Not Available	show chassis power(only total)	Not Applicable since Input is not available
ACX7024(X)(Junos EVO)	show chassis power	show chassis power	Identical
ACX7100(Junos EVO)	show chassis power	show chassis power	Identical
ACX7300(Junos EVO)	show chassis power	show chassis power	Not measured
ACX7500(Junos EVO)	show chassis power	show chassis power	Not measured
QFX5110(Junos)	show chassis env pem	Not Available	Not Applicable since Input is not available
QFX5120(Junos)	show chassis env pem	Not Available	Not Applicable since Input is not available
QFX5130(Junos EVO)	show chassis power	show chassis power	Identical
QFX5200(Junos)	show chassis env pem	Not Available	Not Applicable since Input is not available

Product	Output Power	Input Power	Accuracy Delta of CLI Compared to PDU
QFX5210(Junos)	show chassis env pem	Not Available	Not Applicable since Input is not available
QFX5220(Junos EVO)	show chassis power	show chassis power	Not measured
QFX5230(Junos EVO)	show chassis power	show chassis power	Not measured
QFX5240/5241(Junos EVO)	show chassis power	show chassis power	Identical
QFX5700(E)(Junos EVO)	show chassis power	show chassis power	Not measured

Long story short, when the counters are available, it's pretty accurate and reliable. We didn't have all the combinations of devices and PDU to measure it exhaustively, but when the metrology was available, we verified we can fully trust it.

1.3 Principles and Vocabulary

The devices are hosted in a rack and receive the power from a Power Distribution Unit present in this rack.



It's common for networking device to expose the "Output power", basically the amount of energy exiting the Power Supply Module and feeding the different power buses of the system. This number doesn't represent the power actually consumed by the device or provided by the PDU, but only how much is left after the voltage conversion performed by the different Power Modules (referred as PSMs or PEMs depending on the platform).

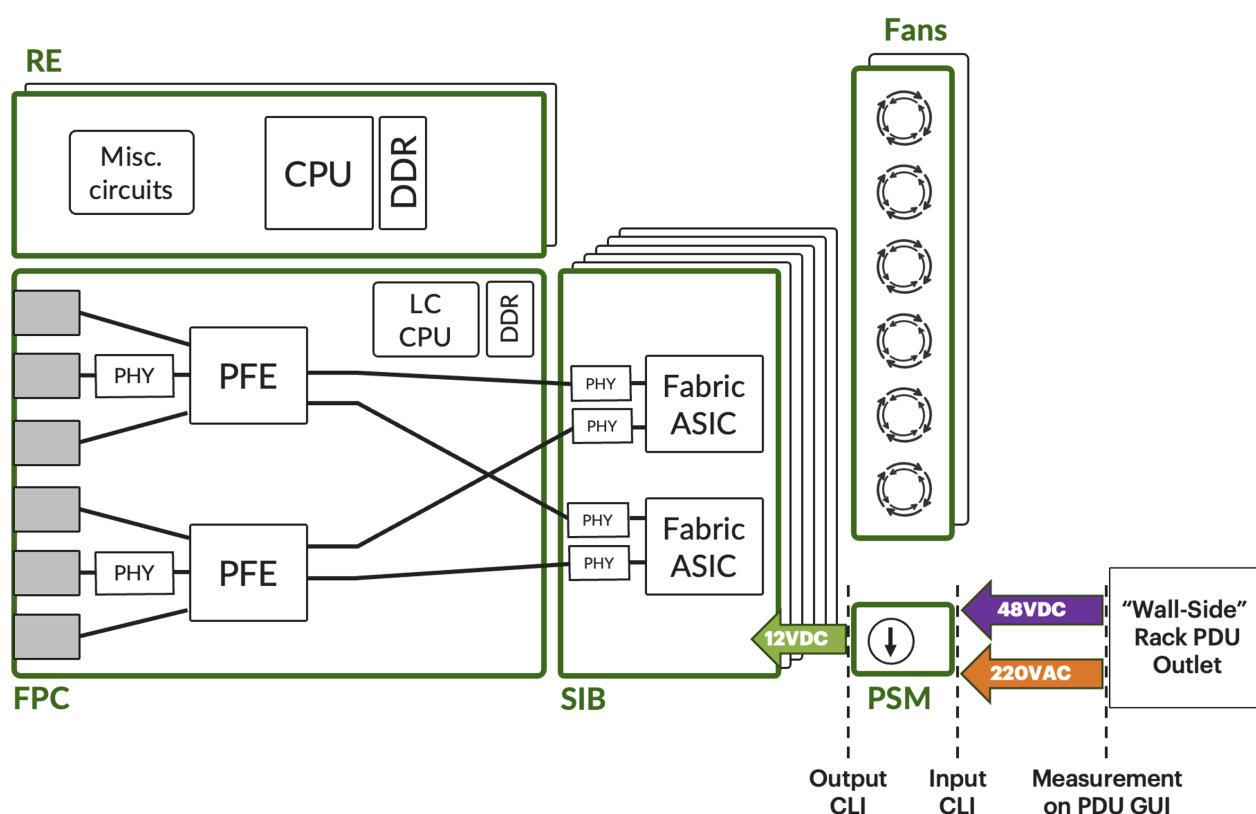
Except for the PTX12000 and legacy MX240/480/960 chassis, most of the routers and switches in the HPE Juniper Networking portfolio are operating with 12V DC out of the power module. This current is converted in different places of the devices (we talk about Point of Load or PoL) to lower voltage: 5V, 3V, 0.8V, etc depending on the components' requirements.

The Output 12V DC current consumption is different from what is injected into the power modules. That's where the concepts of conversion efficiency and ratings are coming from. You know, the "80 Plus" ratings

(Bronze, Silver, Gold, Platinum, Titanium, etc.): it represents the amount of energy lost during the conversion from the Grid to the 12V DC needed by the device. Some products will expose an “Input Power” counter. It represents the voltage, amperage and power received by the Power Module from the grid.

Now the question we will try to answer today is: “*how accurate are these counters and how far can we trust them?*”

The good news is we can precisely monitor the wall-side current in our POC labs. Indeed, the Power Distribution Units (PDUs) can measure the energy they inject into the networking devices, at the outlet level. We will refer to the PDUs’ GUI to collect this information and compare it with the counters on the routers.



The Figure 3 above represents the counters we will collect and compare.

Note: the division of your output and input wattage gives you your conversion ratio/efficiency.

The main objectives of this article are the following:

- Identify the products exposing input power counters
- Clarify the CLI used to expose these counters
- When possible, compare the Input Current in the router CLI to the Outlet output current from the PDU and verify the accuracy

1.4 Disclaimers and Methodology

It's important to understand the following points before proceeding:

- Many factors are influencing the power utilization, starting from temperature, elevation and device utilization. We described these parameters at length in this article PTX10000 Power Optimization <https://juniper.github.io/techposts/ptx10000-power-optimization/article> and video: What does Influence Power Consumption on Networking Devices? <https://www.youtube.com/watch?v=-aEYNDRB9Ck>

- The lab is not a controlled environment and doesn't try to answer the question “*how much is consumed by product X, Y or Z?*” but instead, we want to know “*what's available and how accurate are the power usage counters exposed on product X, Y or Z?*”.
- By nature, power usage is fluctuating. The collection of power consumption counter is just a snapshot of the situation at a given time. That's why we recommend using:

Telemetry and data visualization (Grafana-like dashboards) to draw an average view

If CLI is your only option, collect a dozen of times and calculate an average value. It's not perfect, but still better than just one snapshot number. - And when you'll look at the fluctuating measurements, you'll understand that a margin of error of a couple of watts is the norm, sometimes it varies in a range of 10 Watts. So don't waste time collecting values with decimals.

That's the approach we took for this study, we collected 10 to 12 show commands in less than a minute and we calculated the average value. For the sake of brevity, we will only display one show command in the outputs below, but consider we will calculate the deviation based on an averaged data collection.

1.5 Devices with No Counters For “Input Power”

Let's start listing the boxes that don't expose the input power. Of course, no point trying to verify the accuracy of the measurement.

1.6 MX204

```

1 jnpr@mx204-205> show chassis po
2 ^
3 syntax error.
4 jnpr@mx204-205> show chassis ?
5 Possible completions:
6   alarms Show alarm status
7   ambient-temperature Show chassis ambient temperature
8   environment Show component status and temperature, cooling system speeds
9   errors Show chassis errors
10  fan Show fan and fan tray information
11  firmware Show firmware and operating system version for components
12  fpc Show Flexible PIC Concentrator status
13  fru Show fru replacement status
14  hardware Show installed hardware components
15  hss Show SerDes information
16  location Show physical location of chassis
17  mac-addresses Show media access control addresses
18  memory-usage-chassisd Dump LMLD data to /var/log/chassisd_memory_usage
19  network-services Show chassis network services
20  pic Show Physical Interface Card state, type, and uptime
21  routing-engine Show Routing Engine status
22  satellite-cluster Show Satellite Cluster information
23  synchronization Show clock synchronization information
24  temperature-thresholds Show chassis temperature threshold settings
25 jnpr@mx204-205>
26
27 jnpr@mx204-205> show chassis environment pem
28 PEM 0 status:
29   State Online
30   Airflow Front to Back
31   Temp Sensor 0 37 degrees C / 98 degrees F
32   Temp Sensor 1 35 degrees C / 95 degrees F
33   Firmware version 00.05
34   Fan Sensor 5970 RPM
35   DC Output Voltage(V) Current(A) Power(W) Load(%)
36   11.90 15.00 178 27
37   Health check Information:
38   Status: Unsupported
39 PEM 1 status:
40   State Offline

```

```

41 Health check Information:
42   Status: Unsupported
43
44 jnpr@mx204-205>

```

Only the Output power can be monitored and through the “show chassis env” output.

Input power is not monitored.

1.7 MX240 Chassis

```

1 jnpr@mx240-207-re0> show chassis power detail
2
3 PEM 0:
4   State: Online
5   AC input: OK (1 feed expected, 1 feed connected)
6   Capacity: 2400 W (maximum 2400 W)
7   DC output: 114 W (zone 0, 2 A at 57 V, 4% of capacity)
8
9 PEM 1:
10  State: Present
11  AC input: Out of range (1 feed expected, 1 feed connected)
12  Capacity: 2400 W (maximum 2400 W)
13  DC output: 0 W (zone 0, 0 A at 0 V, 0% of capacity)
14
15 PEM 2:
16  State: Empty
17  Input: Absent
18
19 PEM 3:
20  State: Empty
21  Input: Absent
22
23 System:
24  Zone 0:
25    Capacity: 2400 W (maximum 2400 W)
26    Allocated power: 250 W (2150 W remaining)
27    Actual usage: 114 W
28  Total system capacity: 2400 W (maximum 2400 W)
29  Total remaining power: 2150 W
30
31 jnpr@mx240-207-re0> show chassis environment pem
32 PEM 0 status:
33   State Online
34   Temperature OK
35   DC Output Voltage(V) Current(A) Power(W) Load(%)
36                               57 2 114 4
37 PEM 1 status:
38   State Present
39
40 jnpr@mx240-207-re0>

```

Only the Output power can be monitored and is available in both “show chassis power” and “show chassis env” outputs.

1.8 MX480 Chassis

```

1 jnpr@ASHBVALB-MSE01-AA-IE1> show chassis power detail
2
3 PEM 0:
4   State: Online
5   AC input: OK (1 feed expected, 1 feed connected)
6   Capacity: 2050 W (maximum 2050 W)
7   DC output: 580 W (zone 0, 10 A at 58 V, 28% of capacity)
8
9 PEM 1:
10  State: Online
11  AC input: OK (1 feed expected, 1 feed connected)
12  Capacity: 2050 W (maximum 2050 W)
13  DC output: 627 W (zone 0, 11 A at 57 V, 30% of capacity)
14

```

```

15 PEM 2:
16   State: Present
17   AC input: Out of range (1 feed expected, 1 feed connected)
18   Capacity: 2050 W (maximum 2050 W)
19   DC output: 0 W (zone 0, 0 A at 0 V, 0% of capacity)
20
21 PEM 3:
22   State: Present
23   AC input: Out of range (1 feed expected, 1 feed connected)
24   Capacity: 2050 W (maximum 2050 W)
25   DC output: 0 W (zone 0, 0 A at 0 V, 0% of capacity)
26
27 System:
28   Zone 0:
29     Capacity: 4100 W (maximum 4100 W)
30     Allocated power: 2236 W (1864 W remaining)
31     Actual usage: 1207 W
32   Total system capacity: 4100 W (maximum 4100 W)
33   Total remaining power: 1864 W
34
35 Item Used(W) Max(W)
36 CB 0 + RE 0 242 390
37 CB 0 197 280
38 CB 1 + RE 1 300 510
39 CB 1 252 400
40 FPC 2 283 376
41 FPC 3 275 465
42 FPC 5 300 465
43
44 {master}
45 jnpr@ASHBVALB-MSE01-AA-IE1> show chassis environment pem
46 PEM 0 status:
47   State Online
48   Temperature OK
49   DC Output Voltage(V) Current(A) Power(W) Load(%)
50                               58 10 580 28
51 PEM 1 status:
52   State Online
53   Temperature OK
54   DC Output Voltage(V) Current(A) Power(W) Load(%)
55                               57 11 627 30
56 PEM 2 status:
57   State Present
58 PEM 3 status:
59   State Present
60
61 {master}
62 jnpr@ASHBVALB-MSE01-AA-IE1>

```

Only the Output power can be monitored and is available in both “show chassis power” and “show chassis env” outputs.

1.9 MX960 Chassis

```

1 jnpr@NYCMNY-VFTTP-395_RE0> show chassis power detail
2
3 PEM 0:
4   State: Online
5   AC input: OK (2 feed expected, 2 feed connected)
6   Capacity: 4100 W (maximum 4100 W)
7   DC output: 1026 W (zone 0, 18 A at 57 V, 25% of capacity)
8
9 PEM 1:
10  State: Online
11  AC input: OK (2 feed expected, 2 feed connected)
12  Capacity: 4100 W (maximum 4100 W)
13  DC output: 1824 W (zone 1, 32 A at 57 V, 44% of capacity)
14
15 PEM 2:
16  State: Online
17  AC input: OK (2 feed expected, 2 feed connected)
18  Capacity: 4100 W (maximum 4100 W)
19  DC output: 855 W (zone 0, 15 A at 57 V, 20% of capacity)
20
21 PEM 3:

```

```

22 State: Empty
23 Input: Absent
24
25 System:
26 Zone 0:
27 Capacity: 4100 W (maximum 4100 W)
28 Allocated power: 3090 W (1010 W remaining)
29 Actual usage: 1881 W
30 Zone 1:
31 Capacity: 4100 W (maximum 4100 W)
32 Allocated power: 3155 W (945 W remaining)
33 Actual usage: 1824 W
34 Total system capacity: 8200 W (maximum 8200 W)
35 Total remaining power: 1955 W
36
37 Item Used(W) Max(W)
38 CB 0 + RE 0 351 510
39 CB 0 306 400
40 CB 1 + RE 1 346 510
41 CB 1 302 400
42 CB 2 293 400
43 FPC 1 249 465
44 FPC 2 252 465
45 FPC 3 253 465
46 FPC 4 265 465
47 FPC 5 265 465
48 FPC 8 249 465
49 FPC 9 260 465
50 FPC 10 260 465
51 FPC 11 259 465
52
53 {master}
54 jnpr@NYCMNY-VFTTP-395_RE0> show chassis environment pem
55 PEM 0 status:
56 State Online
57 Temperature OK
58 DC Output Voltage(V) Current(A) Power(W) Load(%)
59 57 19 1083 26
60 PEM 1 status:
61 State Online
62 Temperature OK
63 DC Output Voltage(V) Current(A) Power(W) Load(%)
64 57 32 1824 44
65 PEM 2 status:
66 State Online
67 Temperature OK
68 DC Output Voltage(V) Current(A) Power(W) Load(%)
69 57 15 855 20
70
71 {master}
72 jnpr@NYCMNY-VFTTP-395_RE0>

```

Only the Output power can be monitored and is available in both “show chassis power” and “show chassis env” outputs.

1.10 MX301

```

1 jnpr@mx301-202> show chassis power detail
2
3 PEM 0:
4 State: Online
5 Capacity: 850 W (maximum 850 W)
6 AC input: OK (INP1 feed expected, INP1 feed connected)
7 DC output: 145 W (zone 0, 12.00 A at 12.13 V, 17% of capacity)
8
9 PEM 1:
10 State: Offline
11 Capacity: 850 W (maximum 850 W)
12 AC input: Not ready
13 DC output: 0 W (zone 0, 0.00 A at 0.00 V, 0% of capacity)
14
15 System:
16 Zone 0:
17 Capacity: 850 W (maximum 850 W)
18 Allocated power: 820 W (30 W remaining)

```

```

19 Actual usage: 145 W
20 Total system capacity: 850 W (maximum 850 W)
21 Total remaining power: 30 W
22
23 Item Used(W)
24 Fan Tray 0 1
25 Fan Tray 1 1
26 Fan Tray 2 1
27 Fan Tray 3 1
28 Fan Tray 4 1
29 Fan Tray 5 1
30 RE0/CB0 51
31 FPC 0 99
32
33 jnpr@mx301-202> show chassis environment pem
34 PEM 0 status:
35 State Online
36 Airflow Front to Back
37 Primary Hot Spot 42 degrees C / 107 degrees F
38 Temp Sensor 1 31 degrees C / 87 degrees F
39 Air Outlet ambient 40 degrees C / 104 degrees F
40 Secondary Hot Spot 44 degrees C / 111 degrees F
41 Fan 0 4875 RPM
42 Fan 1 4650 RPM
43 DC Output Voltage(V) Current(A) Power(W) Load(%)
44 12.13 11.50 139 16
45 Health check Information:
46 Status: Unsupported
47 PEM 1 status:
48 State Offline
49 Health check Information:
50 Status: Unsupported
51
52 jnpr@mx301-202>

```

Only the Output power can be monitored and is available in both “show chassis power” and “show chassis env” outputs.

1.11 ACX7020

```

1 regress@rtme-acx7020-01> show chassis power detail
2 Chassis Power Voltage(V) Power(W)
3
4 Total Input Power 47
5 PSM 0
6 State: Online
7 Capacity 220 W (maximum 220 W)
8 PSM 1
9 State: Online
10 Capacity 220 W (maximum 220 W)
11
12 System:
13 Zone 0:
14 Capacity: 440 W (maximum 440 W)
15 Actual usage: 48 W
16 Total system capacity: 440 W (maximum 440 W)
17
18 regress@rtme-acx7020-01> show chassis environment psm
19 PSM 0 status:
20 State Online
21 DC Output OK
22 Health check Information:
23 Status: Health Check Not Supported
24 Last Result: Not Performed
25 Last Execution: Empty
26 Next Scheduled Run: Not Scheduled
27 PSM 1 status:
28 State Online
29 DC Output OK
30 Health check Information:
31 Status: Health Check Not Supported
32 Last Result: Not Performed
33 Last Execution: Empty
34 Next Scheduled Run: Not Scheduled
35

```

```
36 regress@rtme-acx7020-01>
```

The ACX7020 output is pretty unique compared to other Junos or EVO platforms since it displays only the Total input power in the “show chassis power” CLI. But in reality, this number represents the power coming into the hotswap controller after the power supply unit. This number corresponds to the “Output Power” we express on other platforms.

1.12 QFX5110/QFX5120/QFX5200/QFX5210

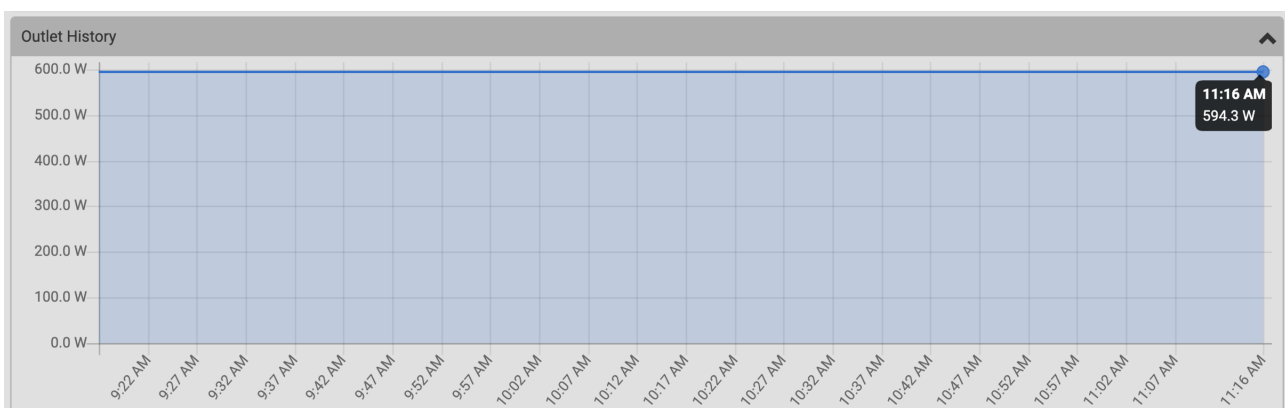
```
1 root@5a7-qfx4> show chassis environment pem
2 localre:
3 -----
4 FPC 0 PEM 0 status:
5   State Present
6 FPC 0 PEM 1 status:
7   State Online
8   Airflow Back to Front
9   Temp Sensor 0 OK 51 degrees C / 123 degrees F
10  Temp Sensor 1 OK 30 degrees C / 86 degrees F
11  Temp Sensor 2 OK 24 degrees C / 75 degrees F
12  Fan 0 4140 RPM
13  Fan 1 3960 RPM
14  DC Output Voltage(V) Current(A) Power(W) Load(%)
15      12 10 120 18
16
17 {master:0}
18 root@5a7-qfx4>
```

Only the Output power can be monitored and through the “show chassis env” output. Input power is not monitored.

All these QFX switches running Junos (non-EVO) have the same CLI output.

1.13 Devices with “Input Power” Counters

1.14 MX304



More or less 594W measured on the PDU

```
1 jnpr@mx304-219-re0> show chassis power detail
2
3 PEM 0:
4   State: Online
5   Capacity: 2200 W (maximum 2200 W)
6   AC input: OK (INPIN feed expected, INPIN1 feed connected)
7   DC output: 556 W (zone 0, 46.19 A at 12.05 V, 25% of capacity)
8
9 PEM 1:
10  State: Offline
```

```

11 Capacity: 2200 W (maximum 2200 W)
12 AC input: Not ready
13 DC output: 0 W (zone 0, 0.00 A at 0.00 V, 0% of capacity)
14
15 System:
16   Zone 0:
17     Capacity: 2200 W (maximum 2200 W)
18     Allocated power: 1845 W (355 W remaining)
19     Actual usage: 556 W
20     Total system capacity: 2200 W (maximum 2200 W)
21     Total remaining power: 355 W
22
23 Item Used(W)
24   Fan Tray 0 4
25   Fan Tray 1 4
26   Fan Tray 2 3
27   SFB 0 98
28   FPC 0 345
29   RE0 46
30   RE1 52
31
32 {master}
33 jnpr@mx304-219-re0> show chassis environment pem
34 PEM 0 status:
35   State Online
36   Airflow Front to Back
37   Inlet Ambient Temp 34 degrees C / 93 degrees F
38   Secondary SR Hotspot 43 degrees C / 109 degrees F
39   Temp Sensor 2 43 degrees C / 109 degrees F
40   Firmware version 1.5, 2.5
41   Fan 0 12992 RPM
42   DC Output Voltage(V) Current(A) Power(W) Load(%)
43                       12.05 45.62 549 24
44   Input Voltage(V) Current(A) Power(W)
45   INP INP1 233.5 2.5 570.0
46   Health check Information:
47     Status: Scheduled
48     Last Result: Not Performed Yet
49     Last Execution: Empty
50     Next Scheduled Run: 2026-05-19 10:41:35
51   PEM 1 status:
52     State Offline
53     Input Voltage(V) Current(A) Power(W)
54     INP INP1 0.0 0.0 0.0
55     Health check Information:
56       Status: Scheduled
57       Last Result: Not Performed Yet
58       Last Execution: Empty
59       Next Scheduled Run: 2026-05-19 10:41:35
60
61 jnpr@mx304-219-re0>

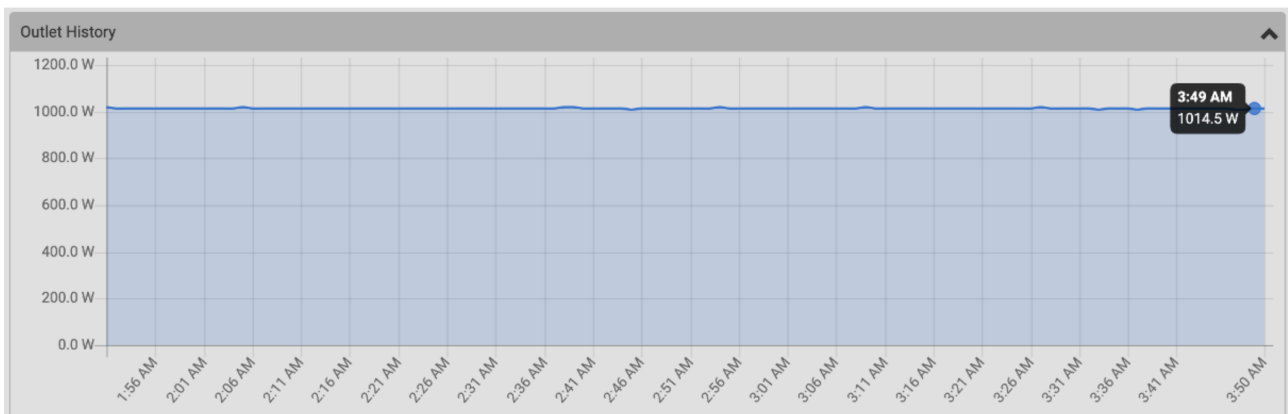
```

Input power is available in the “show chassis env pem”, grep on “INP INP1” to shorten it.

We collected 581.3W in average on the CLI vs 584W approximately on the PDU. That’s 2.2% less on the CLI compared to the rack info.

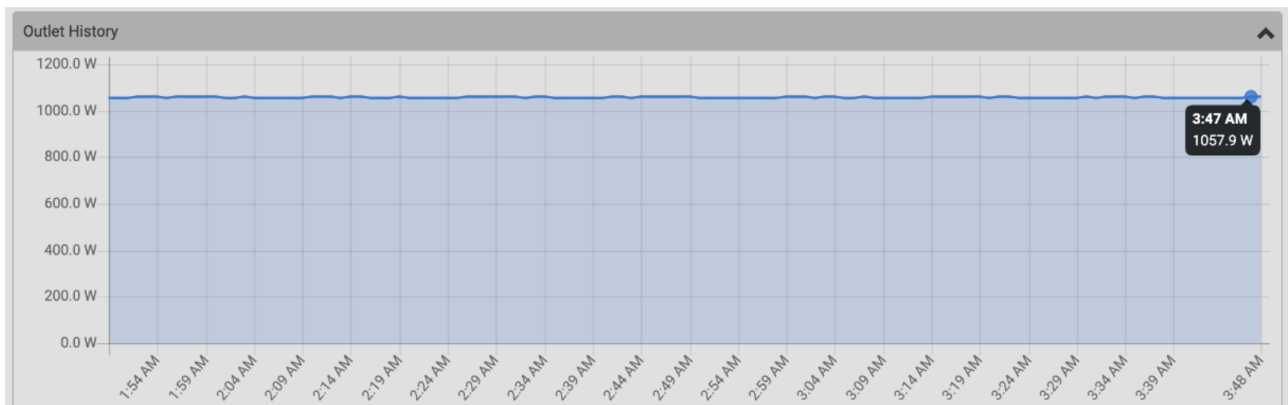
1.15 MX10008 Chassis

PEM0:



from 1014W to 1018W measured on the PDU, let's take a midpoint at 1016W

PEM2:



from 1055W to 1061W measured on the PDU, let's take a midpoint at 1058W

On the CLI:

```

1 jnpr@mx10008-200-re0> show chassis power detail
2
3 PEM 0:
4   State: Online
5   Capacity: 5000 W (maximum 5500 W)
6   AC input: OK (INP1 feed expected, INP1 feed connected)
7   Feed1: AC input
8   DC output: 927 W (zone 0, 75.66 A at 12.26 V, 16% of capacity) (30A input select)
9
10 PEM 1:
11   State: Present
12   Capacity: 0 W (maximum 5500 W)
13   AC input: Not ready
14   Feed1: AC input
15   DC output: 0 W (zone 0, 0.00 A at 0.00 V, 0% of capacity)
16
17 PEM 2:
18   State: Online
19   Capacity: 2800 W (maximum 5500 W)
20   AC input: OK (INP1 feed expected, INP1 feed connected)
21   Feed1: AC input
22   DC output: 962 W (zone 0, 78.74 A at 12.22 V, 17% of capacity) (20A input select)
23
24 PEM 3:
25   State: Present
26   Capacity: 0 W (maximum 5500 W)
27   AC input: Not ready
28   Feed1: AC input
29   DC output: 0 W (zone 0, 0.00 A at 0.00 V, 0% of capacity)
30
31 PEM 4:
32   State: Present
33   Capacity: 0 W (maximum 5500 W)
34   AC input: Not ready
35   Feed1: AC input

```

```

36  DC output: 0 W (zone 0, 0.00 A at 0.00 V, 0% of capacity)
37
38  PEM 5:
39  State: Present
40  Capacity: 0 W (maximum 5500 W)
41  AC input: Not ready
42  Feed1: AC input
43  DC output: 0 W (zone 0, 0.00 A at 0.00 V, 0% of capacity)
44
45  System:
46  Zone 0:
47  Capacity: 7800 W (maximum 7800 W)
48  Allocated power: 7334 W (466 W remaining)
49  Actual usage: 1889 W
50  Total system capacity: 7800 W (maximum 7800 W)
51  Total remaining power: 466 W
52
53  Item Used(W)
54  Fan Tray 0 236
55  Fan Tray 1 225
56  RE0/CB0 90
57  RE1/CB1 92
58
59  Item Used(W) Max(W)
60  SFB 0 123 400
61  SFB 1 120 400
62  SFB 2 125 400
63  SFB 3 129 400
64  SFB 4 123 400
65  SFB 5 125 400
66  FPC 0 517 1770
67
68  jnpr@mx10008-200-re0> show chassis environment pem
69  PEM 0 status:
70  State Online
71  Airflow Front to Back
72  Temp Sensor 0 32 degrees C / 89 degrees F
73  Firmware version AA17AA17.AB15AB15.AC27
74  Fan 0 3968 RPM
75  Fan 1 4640 RPM
76  DC Output Voltage(V) Current(A) Power(W) Load(%)
77  12.26 75.66 927 16
78  Input Voltage(V) Current(A) Power(W)
79  INP 1 228.0 4.6 1053.4
80  INP 2 0.0 0.0 0.0
81  Health check Information:
82  Status: Scheduled
83  Last Result: Not Performed Yet
84  Last Execution: Empty
85  Next Scheduled Run: 2026-05-27 08:40:23
86  PEM 1 status:
87  State Present
88  Input Voltage(V) Current(A) Power(W)
89  INP 1 0.0 0.0 0.0
90  INP 2 0.0 0.0 0.0
91  Health check Information:
92  Status: Scheduled
93  Last Result: Not Performed Yet
94  Last Execution: Empty
95  Next Scheduled Run: 2026-05-27 08:40:23
96  PEM 2 status:
97  State Online
98  Airflow Front to Back
99  Temp Sensor 0 36 degrees C / 96 degrees F
100  Firmware version AA17AA17.AB15AB15.AC27
101  Fan 0 4384 RPM
102  Fan 1 5056 RPM
103  DC Output Voltage(V) Current(A) Power(W) Load(%)
104  12.22 78.74 962 17
105  Input Voltage(V) Current(A) Power(W)
106  INP 1 229.9 4.8 1094.3
107  INP 2 0.0 0.0 0.0
108  Health check Information:
109  Status: Scheduled
110  Last Result: Not Performed Yet
111  Last Execution: Empty
112  Next Scheduled Run: 2026-05-27 08:40:23
113  PEM 3 status:
114  State Present
115  Input Voltage(V) Current(A) Power(W)
116  INP 1 0.0 0.0 0.0

```

```

117 INP 2 0.0 0.0 0.0
118 Health check Information:
119     Status: Scheduled
120     Last Result: Not Performed Yet
121     Last Execution: Empty
122     Next Scheduled Run: 2026-05-27 08:40:23
123 PEM 4 status:
124     State Present
125     Input Voltage(V) Current(A) Power(W)
126     INP 1 0.0 0.0 0.0
127     INP 2 0.0 0.0 0.0
128     Health check Information:
129         Status: Scheduled
130         Last Result: Not Performed Yet
131         Last Execution: Empty
132         Next Scheduled Run: 2026-05-27 08:40:23
133 PEM 5 status:
134     State Present
135     Input Voltage(V) Current(A) Power(W)
136     INP 1 0.0 0.0 0.0
137     INP 2 0.0 0.0 0.0
138     Health check Information:
139         Status: Scheduled
140         Last Result: Not Performed Yet
141         Last Execution: Empty
142         Next Scheduled Run: 2026-05-27 08:40:23
143
144 jnpr@mx10008-200-re0>

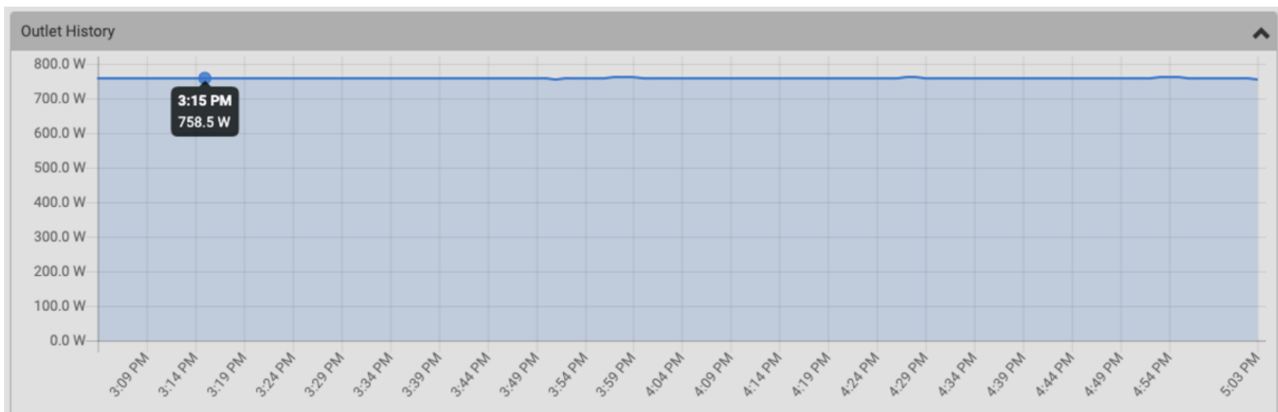
```

Input power is available in the “show chassis env pem”, grep on “INP INP1” to shorten it.

PEM0: We collected 1069W in average on the CLI vs 1016W approximately on the PDU. That’s around 5% more on the CLI compared to the rack info.

PEM2: We collected 1088W in average on the CLI vs 1058W approximately on the PDU. That’s around 3% more on the CLI compared to the rack info.

1.16 PTX10001-36MR



Approximately 758W measured on the PDU

On the CLI:

```

1 jnpr@ptx10001-216> show chassis power detail
2 Chassis Power Voltage(V) Power(W)
3
4 Total Input Power 760
5 PSM 0
6     State: Online
7     Input 1 236 760
8     Output 11.99 700.15
9     Capacity 3000 W (maximum 3000 W)
10 PSM 1
11     State: Offline

```

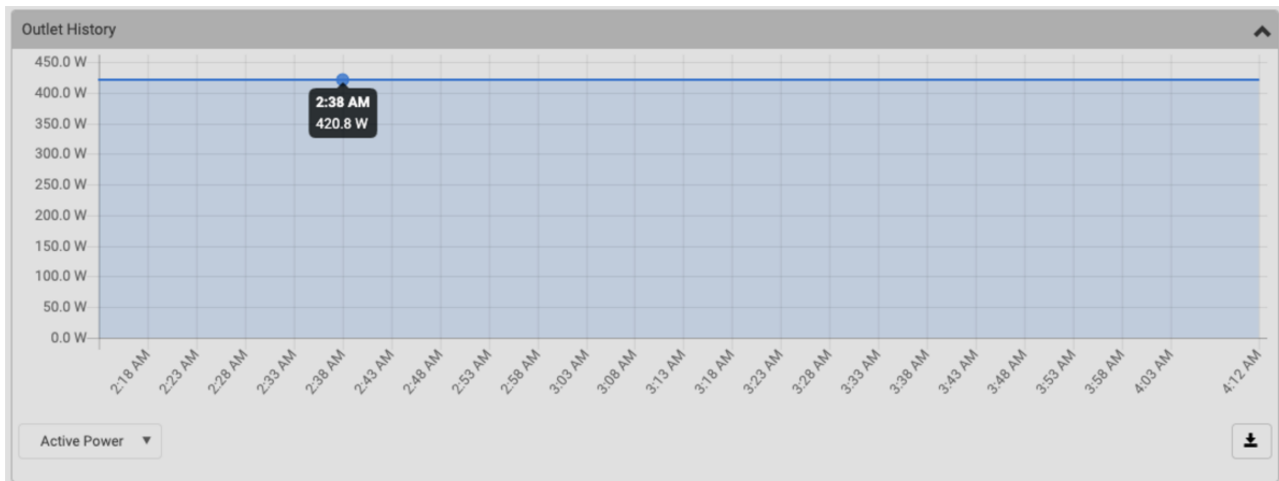
```
12 Input 1 0 0
13 Output 0 0
14 Capacity 0 W (maximum 0 W)
15
16 Item Used(W)
17 Routing Engine 0 34
18 CB 0 5
19
20 System:
21 Zone 0:
22 Capacity: 3000 W (maximum 3000 W)
23 Actual usage: 760 W
24 Total system capacity: 3000 W (maximum 3000 W)
25
26 jnpr@ptx10001-216> show chassis environment psm
27 PSM 0 status:
28 State Online
29 Temperature 33 degrees C / 91 degrees F
30 Fans OK
31 DC Output OK
32 Hours Used 26610
33 Firmware Version 0212.0214
34 Fan 1 14976
35 Fan 2 13248
36 Health check Information:
37 Status: Not Performed
38 Last Result: Not Performed
39 Last Execution: Empty
40 Next Scheduled Run: 2026-05-25 10:52:30 CEST
41 PSM 1 status:
42 State Offline
43 Temperature 31 degrees C / 87 degrees F
44 Fans OK
45 DC Output Failed
46 Hours Used 300
47 Firmware Version 0212.0214
48 Fan 1 0
49 Fan 2 0
50 Health check Information:
51 Status: Not Performed
52 Last Result: Not Performed
53 Last Execution: Empty
54 Next Scheduled Run: 2026-05-25 10:52:40 CEST
55
56 jnpr@ptx10001-216>
```

Both input and output details are provided in the “show chassis power” CLI.

- grep on “Output” for output power
- grep on “Input 1” for input power

We collected 759.8W in average on the CLI vs 758W approx. on the PDU. It’s a marginal difference.

1.17 PTX10002-36QDD



Around 420W measured on the PDU

On the CLI:

```

1  jnpr@ptx10002-204> show chassis power detail
2  Chassis Power Voltage(V) Power(W)
3
4  Total Input Power 416
5    PSM 0
6      State: Online
7      Input 1 233 416
8      Output 12.01 367.91
9      Capacity 3000 W (maximum 3000 W)
10   PSM 1
11     State: Offline
12     Input 1 0 0
13     Output 0 0
14     Capacity 0 W (maximum 0 W)
15
16   Item Used(W)
17     Routing Engine 0 67
18     FPC 0 278
19     Fan Tray 0 1
20     Fan Tray 1 4
21     Fan Tray 2 1
22
23   System:
24     Power mode: Normal
25     Zone 0:
26       Capacity: 3000 W (maximum 3000 W)
27       Allocated power: 2858 W (142 W remaining)
28       Actual usage: 367 W
29     Total system capacity: 3000 W (maximum 3000 W)
30     Total remaining power: 142 W
31
32  jnpr@ptx10002-204> show chassis environment psm
33  PSM 0 status:
34    State Online
35    Temperature 29 degrees C / 84 degrees F
36    Fans OK
37    DC Output OK
38    Hours Used 14391
39    Firmware Version 0212.0215
40    Fan 1 11968
41    Fan 2 10560
42    Health check Information:
43      Status: Not Performed
44      Last Result: Not Performed
45      Last Execution: Empty
46      Next Scheduled Run: 2026-05-22 13:32:49 CEST
47  PSM 1 status:
48    State Offline
49    Temperature 30 degrees C / 86 degrees F
50    Fans OK
51    DC Output Failed

```

```

52  Hours Used 51
53  Firmware Version 0212.0215
54  Fan 1 0
55  Fan 2 0
56  Health check Information:
57      Status: Not Performed
58      Last Result: Not Performed
59      Last Execution: Empty
60      Next Scheduled Run: 2026-05-22 13:32:59 CEST
61
62  jnpr@ptx10002-204>

```

Both input and output details are provided in the “show chassis power” CLI.

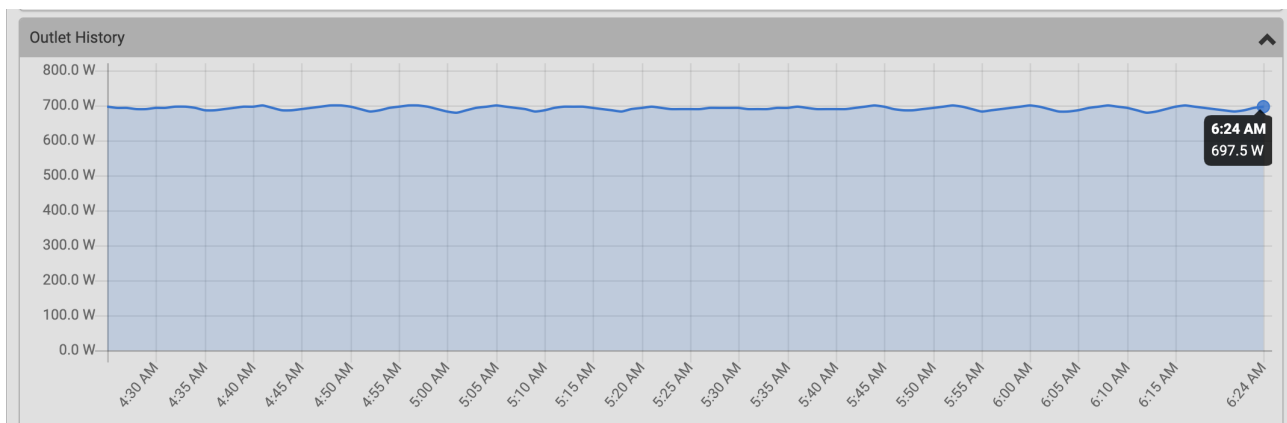
- grep on “Output” for output power
- grep on “Input 1” for input power

We collected 417.2W in average on the CLI vs roughly 420W on the PDU. It’s a marginal difference.

1.18 PTX10000 Chassis

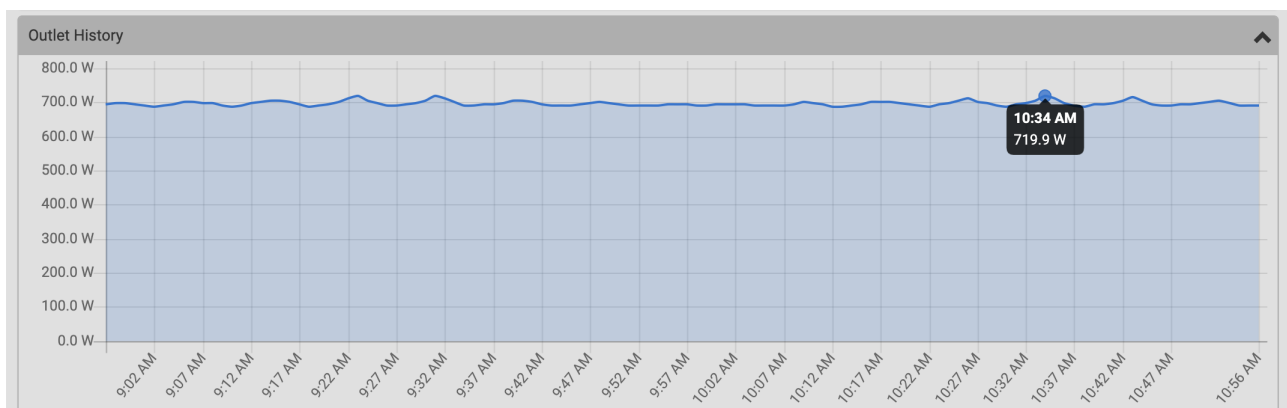
The router in our lab is using 3x PEMs:

PEM0



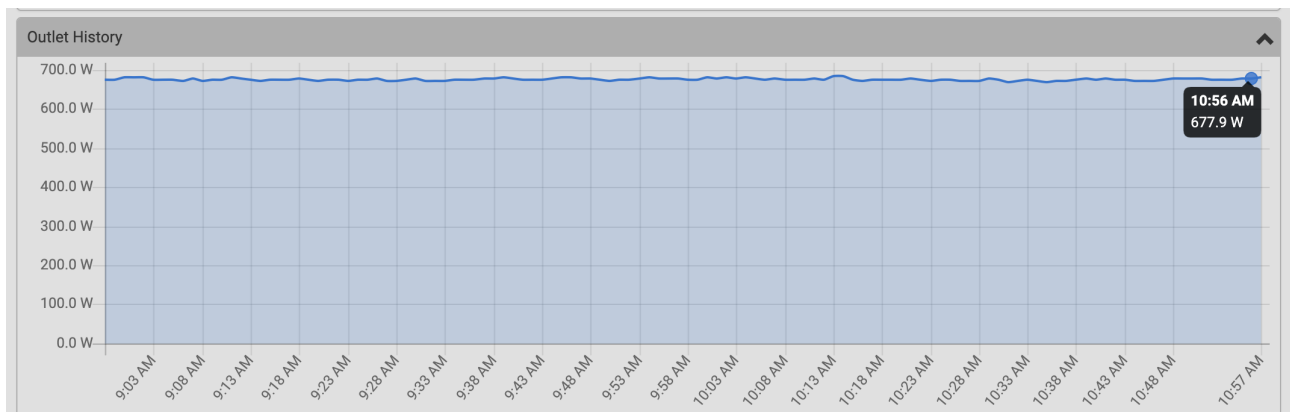
On the PDU, it’s oscillating between 681 and 702W, let’s take a mid-point of 692W for this exercise.

PEM1



Oscillating between 687 and 719W, let’s take a mid-point of 703W for this exercise.

PEM2



Oscillating between 672 and 684W, let's take a mid-point of 678W for this exercise.

On the CLI:

```

1 jnpr@ptx10004-204-re0-re0> show chassis power detail
2 Chassis Power Voltage(V) Power(W)
3
4 Total Input Power 2112
5   PSM 0
6     State: Online
7     INP1 229 696
8     INP2 0 0
9     Output 12.23 608.32 (30A input select)
10    Capacity 5000 W (maximum 5500 W)
11   PSM 1
12     State: Online
13     INP1 231 720
14     INP2 0 0
15     Output 12.21 628.84 (30A input select)
16    Capacity 5000 W (maximum 5500 W)
17   PSM 2
18     State: Online
19     INP1 233 696
20     INP2 0 0
21     Output 12.21 596.6 (30A input select)
22    Capacity 5000 W (maximum 5500 W)
23
24 Item Used(W)
25   CB 0 75
26   FPC 0 817
27   SIB 0 125
28   SIB 1 126
29   SIB 2 125
30   SIB 3 127
31   SIB 4 125
32   SIB 5 125
33   Fan Tray 0 76
34   Fan Tray 1 83
35
36 System:
37   Zone 0:
38     Capacity: 15000 W (maximum 16500 W)
39     Allocated power: 6210 W (8790 W remaining)
40     Actual usage: 1833 W
41     Total system capacity: 15000 W (maximum 16500 W)
42     Total remaining power: 8790 W
43
44 {master}
45
46 jnpr@ptx10004-204-re0-re0> show chassis environment psm
47 PSM 0 status:
48   State Online
49   Temperature 27 degrees C / 80 degrees F
50   Fans OK
51   AC Input1 OK
52   AC Input2 Check
53   Check Input1 Alarm No
54   Check Input2 Alarm Yes
55   DC Output OK
56   Hours Used 33293
57   Firmware Version aa17.aa17.ab15.ab15.ac27
58   Fan 1 5120

```

```
59 Fan 2 6016
60 HVDC Mode Both Inputs AC
61 Health check Information:
62   Status: Not Performed
63   Last Result: Not Performed
64   Last Execution: Empty
65   Next Scheduled Run: 2026-02-18 12:07:00 CET
66 PSM 1 status:
67   State Online
68   Temperature 33 degrees C / 91 degrees F
69   Fans OK
70   AC Input1 OK
71   AC Input2 Check
72   Check Input1 Alarm No
73   Check Input2 Alarm Yes
74   DC Output OK
75   Hours Used 36987
76   Firmware Version aa17.aa17.ab15.ab15.ac27
77   Fan 1 5152
78   Fan 2 6016
79 HVDC Mode Both Inputs AC
80 Health check Information:
81   Status: Not Performed
82   Last Result: Not Performed
83   Last Execution: Empty
84   Next Scheduled Run: 2026-02-18 12:07:00 CET
85 PSM 2 status:
86   State Online
87   Temperature 29 degrees C / 84 degrees F
88   Fans OK
89   AC Input1 OK
90   AC Input2 Check
91   Check Input1 Alarm No
92   Check Input2 Alarm Yes
93   DC Output OK
94   Hours Used 46798
95   Firmware Version aa17.aa17.ab15.ab15.ac27
96   Fan 1 5120
97   Fan 2 6016
98 HVDC Mode Both Inputs AC
99 Health check Information:
100   Status: Not Performed
101   Last Result: Not Performed
102   Last Execution: Empty
103   Next Scheduled Run: 2026-02-18 12:07:00 CET
104
105 {master}
106 jnpr@ptx10004-204-re0-re0>
```

Both input and output information present in the “show chassis power” CLI.

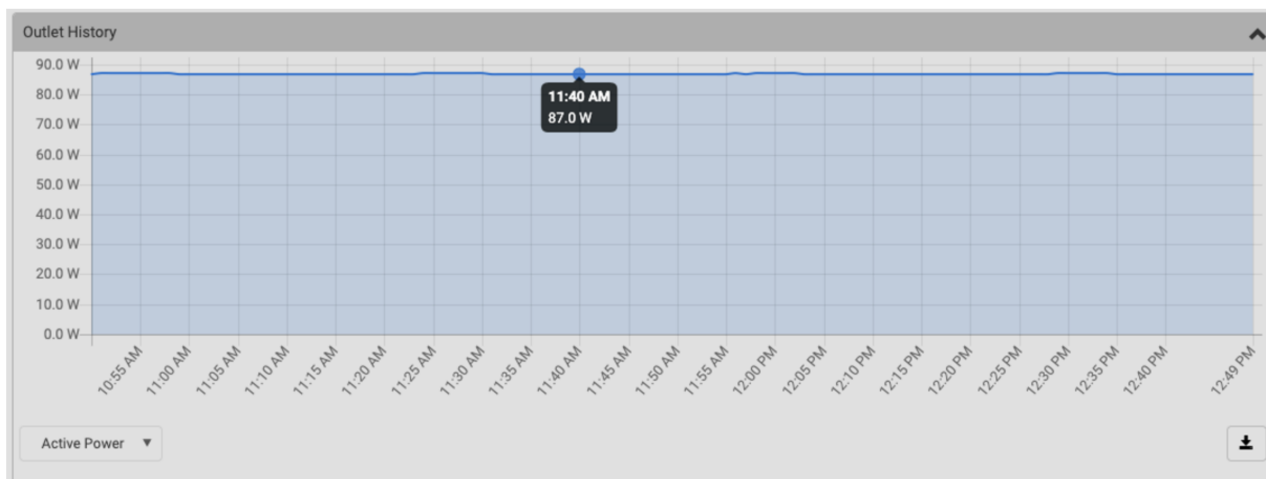
On PEM0, we took a mid-point of 692W for this exercise, compared to 696W, that’s 99%

On PEM1, we took a mid-point of 703W for this exercise, compared to 720W, that’s 98%

On PEM2, we took a mid-point of 678W for this exercise, compared to 696W, that’s 97%.

Considering the constant variation, these calculations don’t make much sense, we can just say the results are pretty much aligned.

1.19 ACX7024



Around 87W measured on the PDU

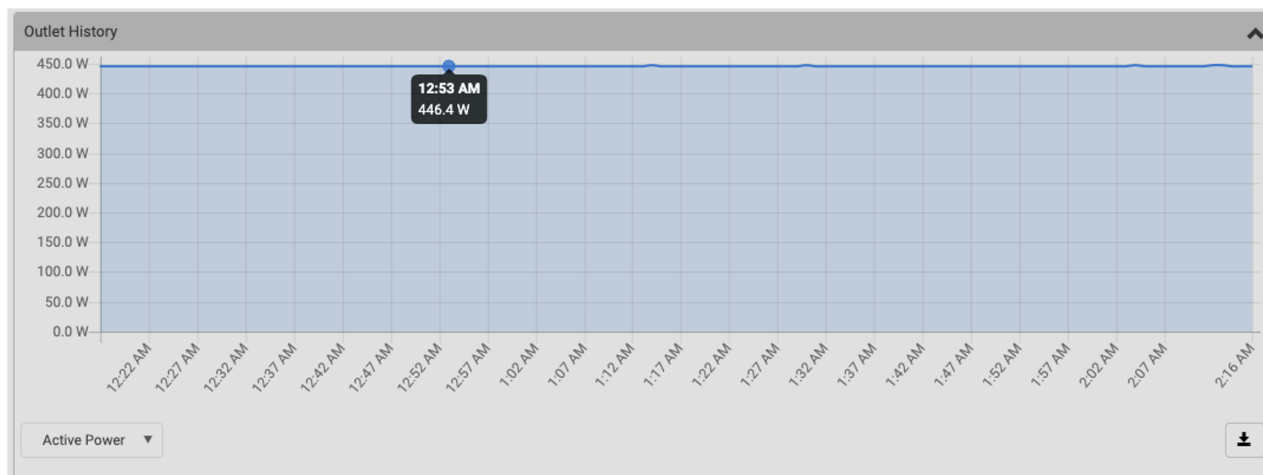
On the CLI:

```
1 jnpr@acx7024-208> show chassis power detail
2 Chassis Power Voltage(V) Power(W)
3
4 Total Input Power 88
5   PSM 0
6     State: Online
7     Input 234 88
8     Output 12.15 75.31
9     Capacity 400 W (maximum 400 W)
10  PSM 1
11    State: Offline
12    Input 0 0
13    Output 0 0
14    Capacity 0 W (maximum 0 W)
15
16 System:
17   Zone 0:
18     Capacity: 400 W (maximum 400 W)
19     Actual usage: 88 W
20     Total system capacity: 400 W (maximum 400 W)
21
22 jnpr@acx7024-208> show chassis environment pem
23 PSM 0 status:
24   State Online
25   Temperature 23 degrees C / 73 degrees F
26   Fans OK
27   DC Output OK
28   Firmware Version 15.17
29   Fan 1 6000
30   Health check Information:
31     Status: Not Performed
32     Last Result: Not Performed
33     Last Execution: Empty
34     Next Scheduled Run: 2026-05-27 17:14:51 CEST
35 PSM 1 status:
36   State Offline
37   Temperature 23 degrees C / 73 degrees F
38   Fans OK
39   DC Output Failed
40   Firmware Version 15.17
41   Fan 1 0
42   Health check Information:
43     Status: Not Performed
44     Last Result: Not Performed
45     Last Execution: Empty
46     Next Scheduled Run: 2026-05-27 17:15:01 CEST
47
48 jnpr@acx7024-208>
```

Both input and output information present in the “show chassis power” CLI. The difference between the

average data collected from CLI (88W) and the current measured on the PDU (87W) is marginal.

1.20 ACX7100



Around 446W measured on the PDU

On the CLI:

```

1 jnpr@acx7100-202> show chassis power detail
2 Chassis Power Voltage(V) Power(W)
3
4 Total Input Power 443
5   PSM 0
6     State: Online
7     Input 231 443
8     Output 11.98 409.1
9     Capacity 1600 W (maximum 1600 W)
10  PSM 1
11    State: Offline
12    Input 0 0
13    Output 0 0
14    Capacity 0 W (maximum 0 W)
15
16 System:
17   Zone 0:
18     Capacity: 1600 W (maximum 1600 W)
19     Actual usage: 409 W
20     Total system capacity: 1600 W (maximum 1600 W)
21
22 jnpr@acx7100-202> show chassis environment pem
23 PSM 0 status:
24   State Online
25   Temperature 30 degrees C / 86 degrees F
26   Fans OK
27   DC Output OK
28   Firmware Version 21.43
29   Fan 1 17984
30   Health check Information:
31     Status: Not Performed
32     Last Result: Not Performed
33     Last Execution: Empty
34     Next Scheduled Run: 2026-05-27 00:23:41 CEST
35 PSM 1 status:
36   State Offline
37   Temperature 35 degrees C / 95 degrees F
38   Fans OK
39   DC Output Failed
40   Firmware Version 00.00
41   Fan 1 0
42   Health check Information:
43     Status: Not Performed
44     Last Result: Not Performed
45     Last Execution: Empty
46     Next Scheduled Run: 2026-05-27 00:23:51 CEST
47

```

```
48 jnpr@acx7100-202>
```

Both input and output information present in the “show chassis power” CLI. The difference between the average data collected from CLI (443.5W) and the current measured on the PDU (446W) is marginal.

1.21 ACX73xx

We didn’t have it behind controlled PDU, we were not able to perform comparison tests.

On the CLI:

```
1 regress@rtme-acx7332-01-re0> show chassis power detail
2 Chassis Power Voltage(V) Power(W)
3
4 Total Input Power 422
5   PSM 0
6     State: Online
7     Input 200 211
8     Output 12.17 167.36
9     Capacity 2200 W (maximum 2200 W)
10  PSM 1
11    State: Online
12    Input 202 211
13    Output 12.2 154.44
14    Capacity 2200 W (maximum 2200 W)
15
16 Item Used(W)
17   Routing Engine 0 44
18   CB 0 18
19   FPC 2 35
20   FPC 3 23
21   Fan Tray 0 6
22   Fan Tray 1 5
23   Fan Tray 2 5
24   Fan Tray 3 4
25   FEB 0 197
26
27 System:
28   Zone 0:
29     Capacity: 4400 W (maximum 4400 W)
30     Allocated power: 1690 W (2710 W remaining)
31     Actual usage: 349 W
32     Total system capacity: 4400 W (maximum 4400 W)
33     Total remaining power: 2710 W
34
35 {master}
36 regress@rtme-acx7332-01-re0> show chassis environment psm
37 PSM 0 status:
38   State Online
39   Temperature 25 degrees C / 77 degrees F
40   Fans OK
41   DC Output OK
42   Hours Used 13803
43   Firmware Version 11.13
44   Fan 1 10144
45   Health check Information:
46     Status: Not Performed
47     Last Result: Not Performed
48     Last Execution: Empty
49     Next Scheduled Run: 2026-05-28 11:24:33 PDT
50 PSM 1 status:
51   State Online
52   Temperature 25 degrees C / 77 degrees F
53   Fans OK
54   DC Output OK
55   Hours Used 15419
56   Firmware Version 11.13
57   Fan 1 9872
58   Health check Information:
59     Status: Not Performed
60     Last Result: Not Performed
61     Last Execution: Empty
62     Next Scheduled Run: 2026-05-28 11:24:43 PDT
63
64 {master}
```

```
65 regress@rtme-acx7332-01-re0>
```

Both input and output current are exposed in the “show chassis power” CLI, but we were not able to test the accuracy of these sensors.

1.22 ACX7500

We didn’t have this router behind controlled PDU, we were not able to perform comparison tests.

On the CLI:

```
1 regress@rtme-acx7509-01-re0> show chassis power detail
2 Chassis Power Voltage(V) Power(W)
3
4 Total Input Power 622
5   PSM 0
6     State: Online
7     Input 1 202 286
8     Output 12.01 226.83
9     Capacity 3000 W (maximum 3000 W)
10  PSM 1
11    State: Online
12    Input 1 201 336
13    Output 12.02 302.13
14    Capacity 3000 W (maximum 3000 W)
15  PSM 2
16    State: Offline
17    Input 1 0 0
18    Output 0 0
19    Capacity 0 W (maximum 0 W)
20  PSM 3
21    State: Offline
22    Input 1 0 0
23    Output 0 0
24    Capacity 0 W (maximum 0 W)
25
26 Item Used(W)
27   CB 0 66
28   FPC 1 102
29   FPC 2 55
30   FPC 5 77
31   Fan Tray 0 17
32   Fan Tray 1 19
33   FEB 0 192
34   FEB 1 0
35
36 System:
37   Zone 0:
38     Capacity: 6000 W (maximum 6000 W)
39     Allocated power: 1730 W (4270 W remaining)
40     Actual usage: 528 W
41     Total system capacity: 6000 W (maximum 6000 W)
42     Total remaining power: 4270 W
43
44 {master}
45 regress@rtme-acx7509-01-re0> show chassis environment psm
46 PSM 0 status:
47   State Online
48   Temperature 27 degrees C / 80 degrees F
49   Fans OK
50   DC Output OK
51   Hours Used 39214
52   Firmware Version 0210.0211
53   Fan 1 10048
54   Fan 2 8896
55   Health check Information:
56     Status: Not Performed
57     Last Result: Not Performed
58     Last Execution: Empty
59     Next Scheduled Run: 2026-06-02 04:21:15 PDT
60 PSM 1 status:
61   State Online
62   Temperature 29 degrees C / 84 degrees F
63   Fans OK
64   DC Output OK
```

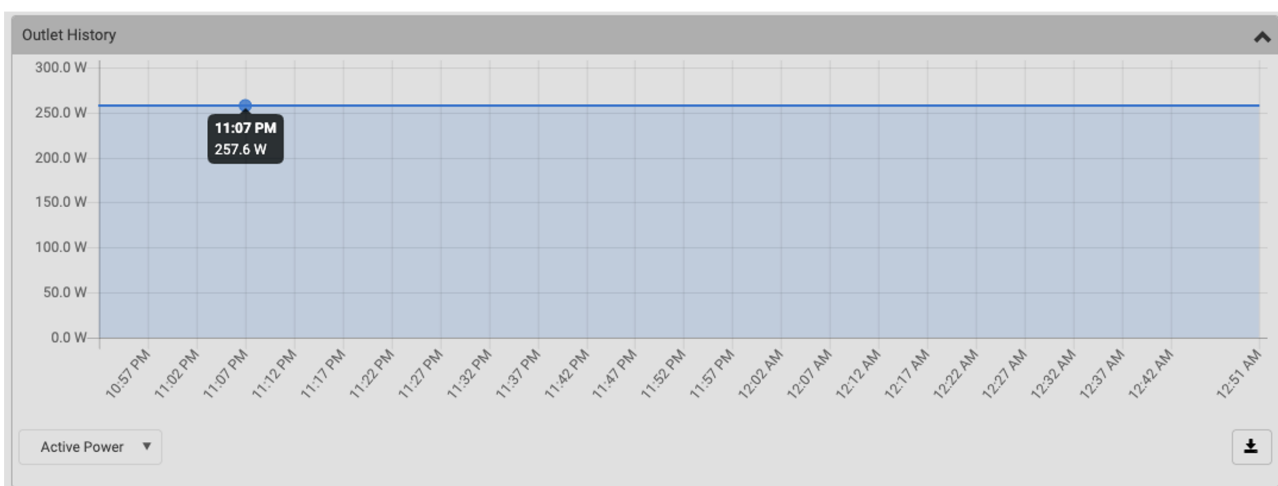
```

65  Hours Used 35646
66  Firmware Version 0210.0211
67  Fan 1 9984
68  Fan 2 8832
69  Health check Information:
70      Status: Not Performed
71      Last Result: Not Performed
72      Last Execution: Empty
73      Next Scheduled Run: 2026-06-02 04:21:25 PDT
74  PSM 2 status:
75      State Offline
76      Temperature 30 degrees C / 86 degrees F
77      Fans OK
78      DC Output Failed
79      Hours Used 541
80      Firmware Version 0212.0214
81      Fan 1 0
82      Fan 2 0
83      Health check Information:
84          Status: Not Performed
85          Last Result: Not Performed
86          Last Execution: Empty
87          Next Scheduled Run: 2026-06-02 04:21:35 PDT
88  PSM 3 status:
89      State Offline
90      Temperature 29 degrees C / 84 degrees F
91      Fans OK
92      DC Output Failed
93      Hours Used 545
94      Firmware Version 0212.0214
95      Fan 1 0
96      Fan 2 0
97      Health check Information:
98          Status: Not Performed
99          Last Result: Not Performed
100         Last Execution: Empty
101         Next Scheduled Run: 2026-06-02 04:21:45 PDT
102
103 {master}
104 regress@rtme-acx7509-01-re0>

```

Both input and output current are exposed in the “show chassis power” CLI, but we were not able to test the accuracy of these sensors.

1.23 QFX5130



Around 257W measured on the PDU

On the CLI:

```

1  jnpr@qfx5130-208> show chassis power detail
2  Chassis Power Voltage(V) Power(W)
3

```

```

4 Total Input Power 256
5   PSM 0
6     State: Online
7     Input 1 233 256
8     Output 11.95 229.79
9     Capacity 1600 W (maximum 1600 W)
10    PSM 1
11      State: Offline
12      Input 1 0 0
13      Output 0 0
14      Capacity 0 W (maximum 0 W)
15
16 System:
17   Zone 0:
18     Capacity: 1600 W (maximum 1600 W)
19     Allocated power: 1241 W (359 W remaining)
20     Actual usage: 229 W
21     Total system capacity: 1600 W (maximum 1600 W)
22     Total remaining power: 359 W
23
24 jnpr@qfx5130-208> show chassis environment pem
25 PSM 0 status:
26   State Online
27   Temperature 28 degrees C / 82 degrees F
28   Fans OK
29   DC Output OK
30   Firmware Version 21.41
31   Fan 1 17952
32   Health check Information:
33     Status: Health Check Not Supported
34     Last Result: Not Performed
35     Last Execution: Empty
36     Next Scheduled Run: Not Scheduled
37 PSM 1 status:
38   State Offline
39   Temperature 36 degrees C / 96 degrees F
40   Fans OK
41   DC Output Failed
42   Firmware Version 00.41
43   Fan 1 0
44   Health check Information:
45     Status: Health Check Not Supported
46     Last Result: Not Performed
47     Last Execution: Empty
48     Next Scheduled Run: Not Scheduled
49
50 jnpr@qfx5130-208>

```

Both input and output current information is available from “show chassis power” CLI.

In average, the CLI collection gives us 256.6W, compared to the 257W we see on the PDU, the difference is marginal.

1.24 QFX5220

On the CLI:

```

1 root@spine1> show chassis power detail
2 Chassis Power Voltage(V) Power(W)
3
4 Total Input Power 420
5   PSM 0
6     State: Online
7     Input 1 202 96
8     Output 12.01 75.4
9     Capacity 1600 W (maximum 1600 W)
10    PSM 1
11      State: Online
12      Input 1 202 97
13      Output 12.05 79.1
14      Capacity 1600 W (maximum 1600 W)
15    PSM 2
16      State: Online
17      Input 1 202 128
18      Output 12.04 109.12

```

```
19 Capacity 1600 W (maximum 1600 W)
20 PSM 3
21 State: Online
22 Input 1 202 99
23 Output 11.99 79.58
24 Capacity 1600 W (maximum 1600 W)
25
26 Item Used(W)
27 Routing Engine 0 47
28 FPC 0 156
29
30 System:
31 Zone 0:
32 Capacity: 6400 W (maximum 6400 W)
33 Allocated power: 1640 W (4760 W remaining)
34 Actual usage: 343 W
35 Total system capacity: 6400 W (maximum 6400 W)
36 Total remaining power: 4760 W
37
38 root@spine1> show chassis environment pem
39 PSM 0 status:
40 State Online
41 Temperature 28 degrees C / 82 degrees F
42 Fans OK
43 DC Output OK
44 Firmware Version 00.00
45 Fan 1 5520
46 Health check Information:
47 Status: Health Check Not Supported
48 Last Result: Not Performed
49 Last Execution: Empty
50 Next Scheduled Run: Not Scheduled
51 PSM 1 status:
52 State Online
53 Temperature 31 degrees C / 87 degrees F
54 Fans OK
55 DC Output OK
56 Firmware Version 00.00
57 Fan 1 5520
58 Health check Information:
59 Status: Health Check Not Supported
60 Last Result: Not Performed
61 Last Execution: Empty
62 Next Scheduled Run: Not Scheduled
63 PSM 2 status:
64 State Online
65 Temperature 29 degrees C / 84 degrees F
66 Fans OK
67 DC Output OK
68 Firmware Version 00.00
69 Fan 1 5456
70 Health check Information:
71 Status: Health Check Not Supported
72 Last Result: Not Performed
73 Last Execution: Empty
74 Next Scheduled Run: Not Scheduled
75 PSM 3 status:
76 State Online
77 Temperature 34 degrees C / 93 degrees F
78 Fans OK
79 DC Output OK
80 Firmware Version 00.00
81 Fan 1 5520
82 Health check Information:
83 Status: Health Check Not Supported
84 Last Result: Not Performed
85 Last Execution: Empty
86 Next Scheduled Run: Not Scheduled
87
88 root@spine1>
```

Both input and output current information is available from “show chassis power” CLI.

We don’t have the proper metrology in our lab PDU to verify the accuracy of the Input counter.

1.25 QFX5230

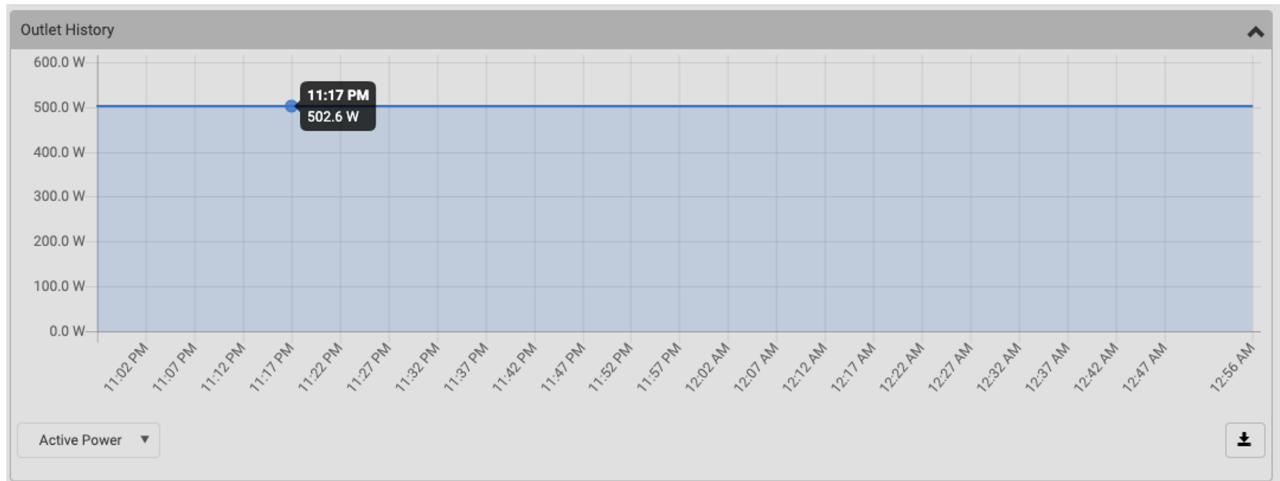
On the CLI:

```
1 root@dc-tme-qfx5230-64cd-01> show chassis power detail
2 Chassis Power Voltage(V) Power(W)
3
4 Total Input Power 304
5   PSM 0
6     State: Online
7     Input 1 204 116
8     Output 11.97 97.32
9     Capacity 3000 W (maximum 3000 W)
10  PSM 1
11    State: Online
12    Input 1 203 188
13    Output 12 169.52
14    Capacity 3000 W (maximum 3000 W)
15
16 Item Used(W)
17   Routing Engine 0 17
18   FPC 0 104
19
20 System:
21   Zone 0:
22     Capacity: 6000 W (maximum 6000 W)
23     Allocated power: 1241 W (4759 W remaining)
24     Actual usage: 266 W
25     Total system capacity: 6000 W (maximum 6000 W)
26     Total remaining power: 4759 W
27
28 root@dc-tme-qfx5230-64cd-01> show chassis environment psm
29 PSM 0 status:
30   State Online
31   Temperature 24 degrees C / 75 degrees F
32   Fans OK
33   DC Output OK
34   Hours Used 20502
35   Firmware Version 0212.0216
36   Fan 1 13568
37   Fan 2 12032
38   Health check Information:
39     Status: Not Performed
40     Last Result: Not Performed
41     Last Execution: Empty
42     Next Scheduled Run: 2026-06-02 18:25:30 PDT
43 PSM 1 status:
44   State Online
45   Temperature 25 degrees C / 77 degrees F
46   Fans OK
47   DC Output OK
48   Hours Used 24800
49   Firmware Version 0212.0216
50   Fan 1 13504
51   Fan 2 12032
52   Health check Information:
53     Status: Not Performed
54     Last Result: Not Performed
55     Last Execution: Empty
56     Next Scheduled Run: 2026-06-02 18:25:40 PDT
57
58 root@dc-tme-qfx5230-64cd-01>
```

Both input and output current information is available from “show chassis power” CLI.

We don’t have the proper metrology in our lab PDU to verify the accuracy of the Input counter.

1.26 QFX5240/5241



Around 502W measured on the PDU

On the CLI:

```

1 jnpr@qfx5240-217> show chassis power detail
2 Chassis Power Voltage(V) Power(W)
3
4 Total Input Power 500
5   PSM 0
6     State: Online
7     Input 1 232 500
8     Output 12.21 437.37
9     Capacity 3000 W (maximum 3000 W)
10  PSM 1
11    State: Offline
12    Input 1 0 0
13    Output 0 0
14    Capacity 0 W (maximum 0 W)
15
16 System:
17   Zone 0:
18     Capacity: 3000 W (maximum 3000 W)
19     Allocated power: 2582 W (418 W remaining)
20     Actual usage: 500 W
21   Total system capacity: 3000 W (maximum 3000 W)
22   Total remaining power: 418 W
23
24 jnpr@qfx5240-217> show chassis environment psm
25 PSM 0 status:
26   State Online
27   Temperature 33 degrees C / 91 degrees F
28   Fans OK
29   DC Output OK
30   Firmware Version 00.02.02
31   Fan 1 7328
32   Health check Information:
33     Status: Health Check Not Supported
34     Last Result: Not Performed
35     Last Execution: Empty
36     Next Scheduled Run: Not Scheduled
37 PSM 1 status:
38   State Offline
39   Temperature 35 degrees C / 95 degrees F
40   Fans OK
41   DC Output Failed
42   Firmware Version 00.02.02
43   Fan 1 0
44   Health check Information:
45     Status: Health Check Not Supported
46     Last Result: Not Performed
47     Last Execution: Empty
48     Next Scheduled Run: Not Scheduled
49
50 jnpr@qfx5240-217>

```

Both input and output current information is available from “show chassis power” CLI.

The average power measured on a dozen of collections from the CLI gives us 500.1W, compared to the 502W we have on the PDU, it's a marginal difference. The counter can be fully trusted.

1.27 QFX5700

On the CLI:

```
1 root@dc-tme-qfx5700-01-re0> show chassis power detail
2 Chassis Power Voltage(V) Power(W)
3
4 Total Input Power 478
5   PSM 0
6     State: Online
7     Input 1 204 220
8     Output 12.01 181.7
9     Capacity 3000 W (maximum 3000 W)
10  PSM 1
11    State: Online
12    Input 1 205 258
13    Output 12.02 219.39
14    Capacity 3000 W (maximum 3000 W)
15
16 Item Used(W)
17   CB 0 52
18   FPC 0 39
19   FPC 1 86
20   FPC 4 68
21   Fan Tray 0 11
22   Fan Tray 1 19
23   FEB 0 128
24
25 System:
26   Zone 0:
27     Capacity: 6000 W (maximum 6000 W)
28     Allocated power: 1690 W (4310 W remaining)
29     Actual usage: 478 W
30   Total system capacity: 6000 W (maximum 6000 W)
31   Total remaining power: 4310 W
32
33 {master}
34 root@dc-tme-qfx5700-01-re0> show chassis environment psm
35 PSM 0 status:
36   State Online
37   Temperature 27 degrees C / 80 degrees F
38   Fans OK
39   DC Output OK
40   Hours Used 21485
41   Firmware Version 0211.0213
42   Fan 1 9984
43   Fan 2 8896
44   Health check Information:
45     Status: Not Performed
46     Last Result: Not Performed
47     Last Execution: Empty
48     Next Scheduled Run: 2026-05-30 17:58:56 UTC
49 PSM 1 status:
50   State Online
51   Temperature 28 degrees C / 82 degrees F
52   Fans OK
53   DC Output OK
54   Hours Used 21576
55   Firmware Version 0211.0213
56   Fan 1 9984
57   Fan 2 8960
58   Health check Information:
59     Status: Not Performed
60     Last Result: Not Performed
61     Last Execution: Empty
62     Next Scheduled Run: 2026-05-30 17:59:06 UTC
63
64 {master}
65 root@dc-tme-qfx5700-01-re0>
```

Both input and output current information is available from “show chassis power” CLI.

We don’t have the proper metrology in our lab PDU to verify the accuracy of the Input counter.

1.28 Conclusion

In this article, we tried to answer a frequent customer question “can I trust the show commands counters related to power usage?”. The answer being “yes, but...”

Yes, but you need to understand the difference between input and output current and understand the platforms supporting “input power”. When this counter is available, for the platforms we were able to test in the lab, the number provided are very reliable.

Also, you certainly noticed the output of “show chassis power” could be very different from one router/switch to another. That’s why a harmonization initiative is ongoing, cross-platform, to provide a unified output. It will be behind “**show chassis power extensive**”. Stay tuned during the next (couple of) releases to see it implemented all over the board.

1.28.1 Useful links

- <https://juniper.github.io/techposts/optimizing-power-consumption-in-high-end-routers/article>
- <https://juniper.github.io/techposts/saving-energy-on-ptx-with-pfe-power-off/article>
- <https://juniper.github.io/techposts/ptx10000-power-optimization/article>
- <https://juniper.github.io/techposts/saving-power-on-acx7000-series/article>
- <https://juniper.github.io/techposts/monitoring-ptx-power-environments-with-telemetry/article>
- <https://juniper.github.io/techposts/ptx12008-power-distribution-and-optimization/article>
- What does Influence Power Consumption on Networking Devices? <https://www.youtube.com/watch?v=-aEYNDRB9Ck>
- Energy Savings in your Network? <https://www.youtube.com/watch?v=TND0GGC0jqk>

1.28.2 Glossary

- AC: Alternating Current
- CB: Control Board
- CLI: Command-Line Interface
- DC: Direct Current
- EVO: Junos Evolved
- FEB: Forwarding Engine Board
- FPC: Flexible PIC Concentrator
- FRU: Field-Replaceable Unit
- GUI: Graphical User Interface
- HPE: Hewlett Packard Enterprise
- HVDC: High-Voltage Direct Current
- INP: Input
- PDU: Power Distribution Unit
- PEM: Power Entry Module

- PFE: Packet Forwarding Engine
- PIC: Physical Interface Card
- PoL: Point of Load
- POC: Proof of Concept
- PSM: Power Supply Module
- PSU: Power Supply Unit
- RE: Routing Engine
- RPM: Revolutions Per Minute
- SFB: Switch Fabric Board
- SIB: Switch Interface Board
- TL;DR: Too Long; Didn't Read

1.28.3 Acknowledgments

Many thanks to the POC team who provided access to their devices for this experiment: Georgios Vlachos and Dejan Tutnjjevic (thanks for your patience Dejan). Thanks to Ridha Hamidi for the help on the QFXs. Thanks to Ankit Gupta for the explanations on ACX7020 architecture.



DC – AI – Routing – Switching – Security – Wireless – Automation

© Copyright 2026 Hewlett Packard Enterprise Development LP. The information contained herein is subject to change without notice. The only warranties for Hewlett Packard Enterprise products and services are set forth in the express warranty statements accompanying such products and services. Nothing herein should be construed as constituting an additional warranty.

Hewlett Packard Enterprise shall not be liable for technical or editorial errors or omissions contained herein.

Trademark acknowledgments, if needed. All third-party marks are the property of their respective owners.

HEWLETT PACKARD ENTERPRISE

HPE.com

